■ Mushroom Business Management & E-Commerce Platform (Mushroom ERP)

Document 01: Project Charter, Product Vision & Strategic Framework

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1. Project Background & Problem Statement

Modern mushroom cultivation in India and global emerging markets represents a high-margin, rapid-cycle agricultural sector. Unlike conventional farming, mushroom cultivation operates in climate-controlled indoor environments with biological turnaround cycles as fast as **18 to 28 days** (for Oyster and Milky mushrooms).

However, commercial mushroom producers face severe operational inefficiencies:
1. **Siloed Record-Keeping:** Production batches, spawn dates, and harvest weights are maintained in paper registers; sales are coordinated via WhatsApp chats; customer ledgers and expenses reside in unformatted Excel files.
2. **Extreme Perishability:** Fresh harvested mushrooms have a shelf life of **24 to 48 hours** at room temperature and 5–7 days under refrigeration. A mismatch between harvest yield and daily retail orders causes immediate product spoilage and severe revenue leakage.
3. **No Batch-to-Profit Visibility:** Producers cannot easily calculate their actual **Cost per Kilogram** because substrate costs, spawn batches, electricity/humidifier expenses, and packaging are never accurately mapped against harvested yields.
4. **Order Double-Booking & Chaos:** Simultaneous orders from retail customers and wholesale restaurants frequently lead to stockouts, cancelled deliveries, and damaged customer trust.

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2. Strategic Solution: The Unified SporeERP Platform

SporeERP unifies the entire enterprise into a single, cohesive **Next.js Full-Stack Application backed by MariaDB**. The system eliminates third-party software sprawl by serving three distinct user tiers through one codebase:

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3. The 2-Founder Lean Business Architecture

The platform is designed around the practical realities of a **2-founder startup**:

4. End-to-End Mushroom Value Chain

Every physical process on the farm is modeled directly into the digital state machine:

5. User Archetypes & Target Personas

6. Strategic Business Metrics & KPI Formulas

The platform embeds mathematical business intelligence into every dashboard:

6.1 Agronomy & Biological KPIs

1. **Biological Efficiency (BE %):**
   $$\text{Biological Efficiency (\%)} = \left(\frac{\text{Total Fresh Mushroom Harvest Weight (kg)}}{\text{Total Dry Substrate Weight (kg)}} \right) \times 100$$

* *Benchmark:* $\ge 80\%$ for Oyster Mushrooms; $\ge 60\%$ for Milky Mushrooms.

2. **Contamination Loss Rate (%):**
   $$\text{Contamination Rate (\%)} = \left(\frac{\text{Number of Contaminated / Discarded Beds}}{\text{Total Inoculated Beds in Batch}} \right) \times 100$$
   * *Target:* $< 3\%$.
3. **Average Yield per Bed (kg/bed):**
   $$\text{Yield per Bed} = \frac{\text{Total Harvested Weight from Batch}}{\text{Total Active Beds in Batch}}$$

6.2 Financial & Commercial KPIs

1. **Unit Production Cost per Kilogram (Rs. /kg):**
   $$\text{Cost per KG} = \frac{\text{Raw Materials (Spawn + Straw)} + \text{Labor} + \text{Utilities (Power + Water)} + \text{Packaging}}{\text{Total Harvested Finished Stock (kg)}}$$
2. **Net Business Profit (Rs.):**
   $$\text{Net Profit} = \text{Gross Revenue} - (\text{COGS} + \text{Operating Expenses} + \text{Logistics} + \text{Taxes})$$
3. **Average Order Value (AOV):**
   $$\text{AOV} = \frac{\text{Total Sales Revenue}}{\text{Total Delivered Orders}}$$
4. **Fresh Stock Spoilage / Wastage Rate (%):**
   $$\text{Spoilage Rate (\%)} = \left(\frac{\text{Weight of Unsold Spoiled Mushrooms (kg)}}{\text{Total Harvested Mushroom Weight (kg)}} \right) \times 100$$
   * *Target:* $< 1.5\%$ (facilitated by automatic dehydration/powder conversion).

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7. Commercial Scaling & SaaS Productization Roadmap

The software architecture is deliberately built with **tenant-ready abstractions** so that once the founders stabilize their own mushroom business, the software can be monetized as a **Vertical SaaS Platform** for thousands of commercial mushroom growers worldwide: